

ECOTEC_Online Student Risk Dataset

Codebook and Metadata Documentation

This codebook describes the structure, variables, metadata, and anonymization procedures associated with the ECOTEC_Online Student Risk Dataset. The repository was organized to support reproducible research in learning analytics, educational data mining, student retention modeling, and responsible artificial intelligence.

Dataset Overview

Institution: Universidad ECOTEC
Country: Ecuador
Educational Modality: Online Higher Education
Number of Records: 12,632
Dataset Type: Institutional tabular dataset
Target Variable: TARGET_RISK

Variable Dictionary

Variable	Description	Type	Missing	Unique Values
PUBLIC_RECORD_ID	Institutional variable: PUBLIC_RECORD_ID	Numeric	0	12632
FACULTY	Academic faculty	Categorical/Text	0	1
DEGREE_PROGRAM	Academic degree program	Categorical/Text	0	29
CURRICULUM_YEAR	Institutional variable: CURRICULUM_YEAR	Numeric	0	4
GENDER	Student gender	Categorical/Text	0	2
AGE	Student age	Numeric	0	62
MARITAL_STATUS	Student marital status	Categorical/Text	0	6
ETHNICITY	Self-reported ethnicity	Categorical/Text	0	6
DISABILITY	Disability condition indicator	Categorical/Text	0	2
INSTITUTION_TYPE	Institutional variable: INSTITUTION_TYPE	Categorical/Text	0	5
NATIONALITY_COUNTRY	Institutional variable: NATIONALITY_COUNTRY	Categorical/Text	0	9
TRANSFER_STUDENT_FLAG	Institutional variable: TRANSFER_STUDENT_FLAG	Categorical/Text	0	2
PAYMENT_PLAN	Institutional payment plan category	Categorical/Text	0	2
ONLINE_ADMISSION_TYPE	Online admission pathway	Categorical/Text	0	15
PAYMENT_DEFERRAL_TYPE	Institutional variable: PAYMENT_DEFERRAL_TYPE	Numeric	0	25
SCHOLARSHIP_NAME	Scholarship category	Categorical/Text	0	9
TARGET_RISK	Student academic risk target variable	Numeric	0	2
REGION	Institutional variable: REGION	Categorical/Text	0	4

Data Anonymization

Personally identifiable information was removed prior to public dissemination. The anonymization workflow excluded names, identification numbers, email addresses, telephone numbers, and direct institutional identifiers. Additional transformations and categorical aggregations were applied to reduce re-identification risk while preserving analytical utility.

FAIR Data Principles

The repository was organized following FAIR data principles to facilitate findability, accessibility, interoperability, and reuse of educational AI resources.

Citation

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